

1. Administrative & Study Identification

Full Study Title: Phase II Randomized, Double-blind, Placebo-controlled Study of LEE011(Ribociclib) or Placebo in Combination With Endocrine Therapy for the Treatment of Pre- and Postmenopausal Chinese Women With HR Positive, HER2-negative, Advanced Breast Cancer, Including a Subset With Pharmacokinetic Analysis **Short Title/Acronym:** Efficacy and Safety of Ribociclib in Pre- and Postmenopausal Chinese Women With HR Positive, HER2-negative, Advanced Breast Cancer [NCT03671330] **Protocol Number:** CLEE011A2206 **NCT Number:** NCT03671330 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ribociclib (LEE011) / Letrozole / Anastrozole / Goserelin (Trade Name: Zoladex, ATC Code: L02AE03) **Phase of Development:** Phase II **Trial Status:** COMPLETED **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, primarily the Progression-Free Survival (PFS), of ribociclib in combination with endocrine therapy compared to placebo plus endocrine therapy. **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), safety, tolerability, and the pharmacokinetics (PK) of ribociclib.

2.2 Background & Rationale To address the need for improved targeted therapies in HR-positive, HER2-negative advanced breast cancer. CDK4/6 inhibitors block the G1/S transition in the cell cycle. This study evaluates the addition of ribociclib to endocrine therapy (letrozole or anastrozole, with goserelin for premenopausal patients) specifically in pre- and postmenopausal Chinese women to demonstrate prolonged progression-free survival and overcome endocrine resistance.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized **Cohorts:** Three cohorts will be enrolled: PK cohort, premenopausal cohort, and postmenopausal cohort.

3.2 Planned Enrollment & Duration Target **\$N\$:** Approximately 327 patients **Number of Sites:** Multiple sites across mainland China **Estimated Study Start:** 29-Aug-2018 **Study Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female patients ≥ 18 years of age. Premenopausal cohort must be < 60 years old at informed consent with a confirmed negative serum pregnancy test or hysterectomy. Postmenopausal cohort must be ≥ 18 years old. 2. Histologically and/or cytologically confirmed diagnosis of HR-positive (estrogen and/or progesterone receptor positive) and HER2-negative (IHC 0, 1+, or 2+ with negative in situ hybridization [FISH, CISH, or SISH]) advanced (locoregionally recurrent or metastatic) breast cancer not amenable to curative therapy. 3. Must have measurable disease per RECIST 1.1 criteria OR, if no measurable disease is present, at least 1 predominantly lytic bone lesion. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 5. Prior therapy allowances: - *Premenopausal cohort*: May have received up to 1 line of chemotherapy for advanced breast cancer (discontinued 28 days before randomization). May have received ≤ 14 days of a NSAI (letrozole or anastrozole) with or without goserelin or goserelin ≤ 28 days prior to randomization. - *Postmenopausal cohort*: Must not have received prior systemic anti-cancer therapy for advanced breast cancer. Patients who received neo(adjuvant) therapy are eligible if the disease-free interval is >12 months from completion. Patients who received ≤ 14 days of letrozole or anastrozole for advanced disease prior to randomization are eligible.

4.2 Exclusion Criteria 1. Prior treatment with any CDK4/6 inhibitor. 2. Known Central Nervous System (CNS) metastases, symptomatic visceral disease, or any disease burden making the patient ineligible for endocrine therapy. 3. Pregnant or nursing (lactating) women, or women of child-bearing potential not using highly effective methods of contraception. Use of oral, transdermal, injected, or implanted hormonal methods of contraception and hormonal replacement therapy (HRT) is strictly prohibited. 4. Unresolved clinical and laboratory acute toxicities related to prior anti-cancer therapy to NCI CTCAE v4.03 Grade ≤ 1 . 5. Known history of Human Immunodeficiency Virus (HIV) infection. 6. Clinically significant, uncontrolled heart disease and/or cardiac repolarization abnormality. 7. Currently receiving any of the prohibited substances as defined in the protocol that cannot be discontinued 7 days prior to the start of treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ribociclib (or matching placebo) given concurrently with a Non-Steroidal Aromatase Inhibitor (NSAI) (Letrozole 2.5 mg daily or Anastrozole 1 mg daily). Premenopausal patients additionally receive Goserelin (Zoladex). **Route of Administration:** Oral (Ribociclib, Letrozole, Anastrozole) and Subcutaneous (Goserelin). **Duration of**

Treatment: 28-day treatment cycles, continuing until disease progression, unacceptable toxicity, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced breast cancer. **Prohibited:** Other systemic anti-cancer therapies (including other hormonal therapy and chemotherapy), other CDK4/6 inhibitors, and concurrent investigational treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Assessment (RECIST 1.1), Labs
Baseline	Day 0	Randomization, First Dose of Study Medication, PK Cohort baseline measures
Interim	Every 4 Weeks	Safety Labs, Treatment Dispensation, PK Evaluation (in PK subset). Tumor imaging every 8-12 weeks
End of Study	Upon Progression	Final Safety Follow-up, Survival Follow-up (OS monitoring)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS). Defined as the time from the date of randomization/start of treatment to the date of the first documented progression or death due to any cause. If a patient has not had an event, PFS is censored at the date of the last adequate tumor assessment.

Measurement Method: Local investigator assessment according to RECIST 1.1 guidelines.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Pharmacokinetic (PK) parameters of Ribociclib, specifically Cmax, Tmax, and AUC0-24h 3. Safety and tolerability profile (Monitoring of AEs/SAEs)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Graded and recorded according to the NCI CTCAE v4.03. Continuous monitoring throughout the treatment period. **Serious Adverse Events (SAEs):** Reported within 24 hours to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC):** Independent oversight committee assigned to routinely evaluate safety signals and trial data integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcome (PFS). Safety Set for all patients who received at least one dose of the study drug. **Sample Size Justification:** Powered to detect a statistically significant improvement in PFS for ribociclib + endocrine therapy versus placebo + endocrine therapy. **Handling of Missing Data:** Standard survival analysis censoring rules apply for PFS and OS; patients without an event are censored at the date of their last adequate tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol compliance and data accuracy. **Audit Trail:** Data entered into validated eCRF systems with comprehensive audit trails and data clarification form mechanisms.

11. Study Identifiers & Governance

Primary ID: CLEE011A2206 **Posting ID:** 556249077819179275 **Registry Number:** NCT03671330 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Efficacy and Safety of Ribociclib in Pre- and Postmenopausal Chinese Women With HR Positive, HER2-negative, Advanced Breast Cancer **Official Regulatory Title:** Phase II Randomized, Double-blind, Placebo-controlled Study of LEE011(Ribociclib) or Placebo in Combination With Endocrine Therapy for the Treatment of Pre- and Postmenopausal Chinese Women With HR Positive, HER2-negative, Advanced Breast Cancer, Including a Subset With Pharmacokinetic Analysis

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: Breast Cancer (Malignant neoplasm of breast) **Preferred Disease Terminology:** Estrogen Receptor Positive Breast Neoplasms, HER2 Negative **Diagnosis Threshold:** ER/PR positive and HER2 negative (IHC 0, 1+, or 2+ with negative in situ hybridization) confirmed by local laboratory. **Medical Methodology:** CDK4/6 inhibitor + Endocrine Therapy combination **Optimization Target:** Inhibition of the G1/S cell cycle transition to prolong disease progression.

13. Study Procedures & Methodology

Management Pathway: Targeted oncology clinical pathway for Advanced Breast Cancer. **Education Protocol (HCP):** Protocol

training on dose modifications/interruptions and safety management (e.g., QT prolongation, neutropenia). **Education Protocol (Patient):** Dosing schedules (21 days on, 7 days off for ribociclib; continuous for NSAI) and strict non-hormonal contraception guidelines. **Quality Audit Mechanism:** Independent central imaging review validation and regular site data verifications.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity. **Onsite Visit Cadence:** Every 4 weeks (Day 1 of each 28-day cycle) **Remote Monitoring Frequency:** Intermittent remote interactions for adverse event screening. **Checklist Review Interval:** Imaging for tumor assessment (RECIST 1.1) every 8-12 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				